

QUESTION-12

Queue Using Linked List

main.c	Output
<pre>1 #include <stdio.h> 2 #include <stdlib.h> 3 struct Node { 4 int data; 5 struct Node *next; 6 }; 7 struct Node *front = NULL; 8 struct Node *rear = NULL; 9 void enqueue(int value) { 10 struct Node *newNode; 11 newNode = (struct Node *)malloc(sizeof (struct Node)); 12 newNode->data = value; 13 newNode->next = NULL; 14 if (rear == NULL) { 15 front = rear = newNode; 16 } else { 17 rear->next = newNode; 18 rear = newNode; 19 } 20 printf("%d inserted into queue\n", value); 21 } 22 void dequeue() { 23 struct Node *temp;</pre>	<pre>10 inserted into queue 20 inserted into queue 30 inserted into queue Queue Elements: 10 20 30 Front Element: 10 Deleted Element: 10 Queue Elements: 20 30 === Code Execution Successful ===</pre>
<pre>24 if (front == NULL) { 25 printf("Queue Underflow\n"); 26 return; 27 } 28 temp = front; 29 printf("Deleted Element: %d\n", front ->data); 30 front = front->next; 31 if (front == NULL) 32 rear = NULL; 33 free(temp); 34 } 35 void peek() { 36 if (front == NULL) 37 printf("Queue is Empty\n"); 38 else 39 printf("Front Element: %d\n", front ->data); 40 } 41 void display() { 42 struct Node *temp; 43 if (front == NULL) { 44 printf("Queue is Empty\n"); 45 return;</pre>	<pre>10 inserted into queue 20 inserted into queue 30 inserted into queue Queue Elements: 10 20 30 Front Element: 10 Deleted Element: 10 Queue Elements: 20 30 === Code Execution Successful ===</pre>